

SRS

Introduction

Purpose: The purpose of this Software Requirements Specification (SRS) is to define the requirements of the **Smart Scientific Calculator**.

The system will provide users with a simple interface for performing basic arithmetic, advanced mathematical, and scientific calculations. The system will be developed using the **Incremental Model**, where functionality will be implemented and tested in multiple increments.

Problem Statement

Users frequently need to perform different types of mathematical calculations, ranging from simple arithmetic operations to scientific calculations.

A calculator application can provide these operations through a single interface. However, instead of developing all functionality at once, this project will develop the calculator progressively using the **Incremental Model**.

Objectives

1. To develop an easy-to-use calculator application.
2. To allow users to perform basic arithmetic operations.
3. To provide advanced mathematical operations.
4. To provide commonly used scientific functions.
5. To handle invalid mathematical operations appropriately.
6. To provide calculation history.
7. To develop the system incrementally.
8. To test each increment before integrating the next increment.

9. To maintain a modular structure that allows additional functionality to be added easily.

Stakeholder Identification :

Stakeholder	Requirement/Interest
End User	Wants to perform mathematical calculations easily and correctly
Project Team	Responsible for analysing, designing, implementing and testing the system
Tester	Verifies that the system produces correct results
Faculty/Supervisor	Evaluates the project and its software-engineering process

User Needs :

UN-01 — Basic Calculation

The user needs to perform common arithmetic calculations.

UN-02 — Advanced Calculation

The user needs operations such as square, square root and power.

UN-03 — Scientific Calculation

The user needs commonly used scientific functions such as trigonometric and logarithmic operations.

UN-04 — Error Handling

The user needs the calculator to identify invalid operations instead of producing misleading results.

UN-05 — Easy Interaction

The user needs a simple interface through which calculations can be entered and results can be viewed.

UN-06 — Calculation History

The user may need to review previously performed calculations.

Scope of the System :

In-Scope Features

The calculator will include:

Basic Operations

- **Addition**
- **Subtraction**
- **Multiplication**
- **Division**
- **Modulus**
- **Percentage**

Advanced Operations

- **Square**
- **Power**
- **Square root**
- **Reciprocal**

Scientific Operations

- **Sine**
- **Cosine**
- **Tangent**
- **Logarithm**
- **Natural logarithm**

Utility Features

- **Clear**
- **Delete/backspace**
- **Calculation history**
- **Error handling**

Out-of-Scope Features

The following features are not included in the current project scope:

- **User registration and authentication**
- **Online/cloud synchronization**
- **Payment functionality**
- **AI-based mathematical problem solving**
- **Online tutoring**
- **Collaborative calculations**
- **Complex graph plotting**
- **Multi-user accounts**

These features considered as future enhancements.

User Interface Requirements

The calculator interface shall provide:

- 1. A display area for entering expressions.**
- 2. A display area or mechanism for showing results.**
- 3. Buttons/controls for numerical input.**
- 4. Buttons/controls for arithmetic operations.**
- 5. Controls for scientific operations.**
- 6. A clear button.**
- 7. A delete/backspace button.**
- 8. Access to calculation history.**

System Constraints

The project has the following constraints:

- 1. The system must be developed within the academic project timeline.**
- 2. Development must follow the selected Incremental Model.**
- 3. Each increment should be tested before proceeding to subsequent functionality.**
- 4. The implementation should remain understandable to all project members.**
- 5. The project must provide documentation and evidence of development progress.**

Assumptions

The following assumptions are made:

- 1. The user has basic knowledge of numerical calculations.**
- 2. The user provides valid numerical input for normal calculations.**
- 3. The application is used for normal mathematical calculations rather than high-precision scientific research.**

- 4. The calculator operates within the numerical limitations of the selected programming environment.**
- 5. Calculation history does not require cloud synchronization in the current version.**